Research Project Report

[Project Title]

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Abstract

Abstract goes here.

Declaration

Student

I hereby declare that:

1. This report is the result of the final year project work carried out by my project partner (see cover page) and I under the guidance of our supervisor (see cover page) in the 2025 academic year at the Department Engineering Science and Biomedical Engineering, Faculty of Engineering, University of Auckland.
2. This report is not the outcome of work done previously.
3. This report is not the outcome of work done in collaboration, except that with a potential project sponsor (if any) as stated in the text.
4. This report is not the same as any report, thesis, conference article or journal paper, or any other publication or unpublished work in any format.

In the case of a continuing project, please state clearly what has been developed during the project and what was available from previous year(s):

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The word count of this report is: < DELETE AND INSERT WORD COUNT>

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Glossary of Terms

Term	Definition

Abbreviations

AOA	Angle of attack

1. Report Template

text

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1.1.1. Subsubsection

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2. Example Section

2.1. Intext References

Introduction

3. Literature Review

4. Methodology

4.1. Research design and evidence boundary

This study used controlled computational-fluid-dynamics comparisons to examine how inlet loading and selected droplet and wall-film treatments affected predicted liquid carryover in a Purnanto-type geothermal separator. The work began from a reproduced reference configuration, then introduced a spatially separated steam-liquid inlet, discrete-particle tracking and Eulerian wall-film mechanisms in staged case families. Formal comparisons were distinguished from exploratory mechanism-screening cases.

The simulations are reported as exploratory model observations rather than validated separator-performance predictions. This claim boundary is stated once here and revisited in the final discussion: the values are useful for identifying model sensitivities and recurring behaviour, but should not be interpreted as field efficiency without stronger mass closure and matched comparisons.

4.2. Reference and project models

My project partner reproduced the Purnanto enthalpy-dependent steam-separation workflow using ANSYS Fluent 2024 R2, the supplied Purnanto mesh and a steady pressure-based Mixture model. Six published phase-flow conditions were defined, with enthalpy used only to identify a prescribed steam-liquid mass split because the Energy equation was disabled. A nine-bin droplet distribution was reconstructed from the Harwell correlation and Purnanto's digitised cumulative distribution, and the same automated carrier-flow, DPM and export sequence was applied to each condition.

The project model retained the spiral-inlet separator geometry and used the mesh and physical settings summarised in Table 1. The original mixed inlet was replaced by a narrow wall-side liquid inlet and a larger inner steam inlet so that phase distribution could be changed without altering the vessel geometry.

Table 1. Principal model settings used in the reported project cases.

Item	Setting
Solver and phases	Steady pressure-based Mixture model; steam primary, liquid water secondary
Turbulence	RNG k-epsilon with standard wall functions
Thermal treatment	Energy disabled; no flashing
Gravity	9.81 m/s² in the negative vertical direction
Project mesh	7,601,261 cells; 1,309,312 nodes
Project inlet	Separate mass-flow inlets for steam and liquid
Steam outlet	Pressure outlet; 1.12 MPa gauge (operating pressure 0 Pa)
Lower boundary	Closed stationary wall

Mesh and Inlet figure goes here!!!!

Figure 1. Separator geometry, computational boundaries and project split inlet.

4.3. Simulation families

The main carrier-flow comparison changed only the inlet phase mass flows while retaining the project geometry, mesh, inlet areas and model stack. Subsequent particle comparisons isolated deterministic and stochastic DPM treatment. The mass-partitioned 5% DPM wall-film family then divided the total liquid between Eulerian and Lagrangian representations before adding wall-film deposition, splash, edge separation, stripping and global DPM interaction in stages.

Table 2. Evidential roles of the simulation families.

Case family	Deliberate change	Role in report
Purnanto six-condition replication	Reported phase split and Harwell-derived droplet distribution	Reference-workflow status
Low/reference/high split inlet	Total inlet loading	Primary carrier-flow comparison
Deterministic/stochastic DPM	Turbulent particle-dispersion treatment	Controlled model-sensitivity comparison
Mass-partitioned two-way DPM	Eulerian/Lagrangian liquid allocation	Method-development result
Clean and mechanism-specific EWF	Deposition, splash, edge separation or stripping	Exploratory mechanism screening
Combined EWF interaction off/on	Global DPM feedback	Provisional coupled comparison

For the mass-partitioned 5% DPM case, the total liquid input was preserved as:

$$\dot{m}_{DPM} = f_{DPM} \times 116.92\ \ kg/s, \qquad \dot{m}_{Eulerian,l} = \left(1 - f_{DPM}\right) \times 116.92\, kg/s \quad (1)$$

The reported mass-partitioned branch used a 5% DPM allocation as a diagnostic sensitivity point rather than a measured geothermal mist fraction.

4.4. Data extraction and reported quantities

Automated Fluent readback and post-processing were used to preserve case lineage, compare parent and child settings and extract phase fluxes, DPM fates and wall-film quantities consistently. The supporting repository links literature evidence, project decisions, setup records and executable PyFluent routines so that model changes can be traced and checked rather than reconstructed from memory.

The carrier-flow output was reported as resolved steam-outlet liquid flux and outlet dryness. Because the project carrier model had no active liquid discharge, the complement of the carryover ratio was not reported as whole-separator efficiency.

$$C_{steam} = \frac{\dot{m}_{\left(l,\ steam\ \ outlet\right)}}{\dot{m}_{\left(l,inlet\right)}} \quad (2)$$

$$x_{out} = \frac{\dot{m}_{\left(v,steam\ \ outlet\right)}}{\dot{m}_{\left(v,steam\ \ outlet\right)} + \dot{m}_{\left(l,steam\ \ outlet\right)}} \quad (3)$$

DPM outcomes were grouped as escaped, trapped, absorbed or incomplete, depending on the active model. For the EWF cases, the reported quantities included film inventory, maximum and area-averaged film thickness, derived mean film speed and generated splash, edge-separation or stripping events. Event counts were kept separate from represented secondary-particle mass.

5. Results

5.1. Reference-replication status

Two of the six Purnanto operating conditions had completed the automated workflow at the reporting snapshot. Table 3 summarises their DPM fate accounting and provisional outlet-quality calculation.

Table 3. Interim outcomes from the completed Purnanto replication conditions.

Condition	Purnanto (%)	Ours (%)	Difference
1600 kJ/kg -25%	99.9915	99.7746	-0.2169 pp
1440 kJ/kg	99.7748	99.6717	-0.1031 pp
1520 kJ/kg	99.8121	99.7304	-0.0817 pp

For both completed conditions, total reported fate mass agreed with injected liquid within 0.2%, but more than two-fifths of the liquid remained incomplete. With four conditions unfinished, no enthalpy-dependent trend was reported.

5.2. Inlet loading and steam-outlet liquid flux

Steam-outlet liquid flux increased monotonically, but the response was strongly uneven: the reference case was approximately 62 times the low-loading case, whereas the high-loading case was about 38% above the reference case.

Table 4. Split-inlet loading inputs and resulting steam-outlet liquid flux.

Condition	Nominal velocity	Liquid inlet	Vapour inlet	Steam-outlet liquid	Outlet Dryness
Low	20.00 m/s	86.18 kg/s	60.21 kg/s	0.00132696 kg/s	99.9978%
Reference	27.1 m/s	116.92 kg/s	80.69 kg/s	0.0821320 kg/s	99.8993%
High	32.14 m/s	138.48 kg/s	96.76 kg/s	0.1132807 kg/s	99.8841%

Mixture imbalance changed little across the loading sweep and remained large in every case. The comparison is therefore reported as a scoped steam-line flux response rather than as whole-separator liquid-removal efficiency.

5.3. Sensitivity to DPM dispersion treatment

Stochastic tracking increased completed escape for the finest class by quite a significant margin relative to deterministic tracking. The highest completed escape range occurred in the 10 μm stochastic cases, while no completed steam-outlet escape was recorded for the larger reported classes.

Table 5. Completed droplet escape fractions under deterministic and stochastic tracking.

Droplet size	Deterministic	Stochastic (Eddy life time on to off)
5 to 5.63 μm	0.37%	10.65% to 12.54%
10 μm	0.046%	13.56% to 15.53%
28.14 μm	No completed escape	No completed escape
40 μm	No completed escape	No completed escape

5.4. Mass partition and EWF mechanism screening

The mass-partitioned case preserved the original total liquid input by reducing the Eulerian inlet by the amount assigned to DPM parcels. The subsequent clean EWF control enabled deposition and drainage while disabling splash, edge separation, stripping and global DPM interaction.

Across the clean, splash, edge-separation, stripping and combined EWF branches, the droplet response followed the same size-ordered pattern. Fine droplets retained a direct steam-outlet escape pathway, intermediate droplets showed increasing wall-film absorption, and the 168.81 µm and larger classes were dominated by absorption. The edge-separation branch reported 120 separated-particle events for the 348.88 µm class. The combined branch reported 20 splash events and 120 separation events. No represented secondary-particle mass was available for these event counts.

Table 6. Final film quantities from the EWF mechanism-screening branches.

EWF branch	Film inventory	Maximum thickness	Area-average thickness
Clean deposition control	0.07150 kg	0.152 mm	1.211 µm
Splash only	0.07431 kg	0.164 mm	1.259 µm
Edge separation only	0.05639 kg	0.123 mm	0.955 µm
Stripping only	0.05440 kg	0.121 mm	0.921 µm
Combined mechanisms	0.05668 kg	0.125 mm	0.960 µm

5.5. Global DPM interaction in the combined EWF model

Enabling global DPM interaction increased film inventory, both thickness measures and derived mean film speed. Inventory and thickness changed by the same proportion, while film speed increased less strongly (Table 7).

Table 7. Captured EWF quantities with global DPM interaction off and on.

Quantity	Interaction off	Interaction on	Difference
Film inventory	0.05668 kg	0.07991 kg	+41%
Maximum thickness	0.125 mm	0.177 mm	+41%
Area-average thickness	0.960 µm	1.354 µm	+41%
Derived mean film speed	0.0533 m/s	0.0684 m/s	+28%
Direct escape share	31.5%	29.3%	−2.1 percentage points
Film absorption share	66.9%	68.1%	+1.2 percentage points
Bottom trapping share	1.65%	2.55%	+0.9 percentage points
Coarse-class splash events	20	44	+24
Coarse-class edge-separation events	120	188	+68
Coarse-class stripping events	0	7	+7

The interaction-on state also produced more coarse-class splash and edge-separation events and introduced stripping events. Bulk steam-outlet vapour flow differed by only approximately 0.002% between the captured states; event counts remain distinct from represented secondary-particle mass.

Bar chart or plot goes here for easy comparison

Figure 2. Summary of the carrier-loading, DPM-treatment and EWF observations.

6. Discussion

6.1. Answer to the research question

The current results show that the predicted separator response is sensitive both to inlet loading and to the selected droplet and wall-interaction treatment. The split-inlet carrier model produced a large increase in resolved steam-outlet liquid flux from low to reference loading, followed by a smaller increase from reference to high loading. The DPM and EWF studies further showed that the predicted pathway of liquid depended strongly on turbulent dispersion, Eulerian-Lagrangian mass allocation and whether wall deposition and feedback were active.

6.2. Interpretation of the loading response

Because the low-loading flux was close to zero, a high-to-low ratio exaggerates the apparent scale of the response. The concentration of the increase between low and reference loading is more consistent with a threshold-like change in the resolved outlet pathway than with a proportional performance law. A graph of absolute flux, preferably also shown on a logarithmic scale, is therefore more informative than a fitted curve.

The carrier-flow comparison cannot determine whole-separator collection efficiency because the lower boundary is closed and the imbalance remains large. The common domain closure still permits a cautious comparison of the resolved steam-line pathway, but not an interpretation of the unreported liquid as successfully collected.

6.3. Interpretation of DPM and wall-film sensitivity

The more-than-order-of-magnitude change in fine-droplet escape between deterministic and stochastic tracking demonstrates that the predicted grade response is model dependent. Stochastic tracking is not automatically more accurate; rather, the comparison shows that turbulent-dispersion settings must be fixed before other case differences can be attributed to loading or wall physics. The lack of completed escape for the larger tested droplets is consistent with stronger inertial separation, but the large incomplete trajectory population prevents a complete grade-efficiency interpretation.

The recurring EWF crossover from fine-droplet escape to coarse-droplet absorption is the strongest family-wide wall-film observation because it appeared across several mechanism branches. However, splash, edge-separation and stripping event counts alone do not quantify re-entrained liquid mass. The combined interaction-on snapshot also showed a thicker and faster local film with slightly more absorption and trapping, but this comparison remains provisional until the off/on cases are matched at the same physical time and initial film state.

6.4. Strengths, limitations and alternative explanations

A strength of the project is the structured simulation lineage: major physics were introduced in stages, and automated readback and post-processing reduced unnoticed configuration drift. The explicit liquid partition in the mass-partitioned wall-film branch also corrected the earlier risk of representing the same liquid simultaneously as a full Eulerian inlet and as additional coupled DPM mass.

The main limitations are the open liquid balance in the split-inlet carrier cases, incomplete DPM trajectories, inferred droplet-bin reconstruction, uncertain equivalence of some captured EWF states and the absence of direct field validation. Alternative explanations for the observed changes therefore include numerical trajectory treatment, different saved-state timing and local film accumulation, rather than only physical separator response. These alternatives should remain visible when interpreting the current patterns.

6.5. Implications and priority next work

The results indicate that increasing model complexity changes not only the magnitude of carryover but also the pathway assigned to the liquid: direct escape, wall absorption, bottom trapping or unresolved residence. For future separator design studies, model-selection sensitivity may therefore be as important as changing geometry or operating condition.

The priority next work is to complete the six-condition Purnanto sweep, localise incomplete particle tracks, establish a physically meaningful liquid-discharge or transient inventory treatment, and rerun matched EWF off/on comparisons from a common initial state and physical time. Generated splash, separation and stripping events should also be converted into represented secondary-particle mass before they are used to quantify re-entrainment.

7. Conclusion

The split-inlet simulations showed a strong but nonlinear sensitivity of resolved steam-outlet liquid flux to inlet loading. DPM and EWF outcomes were also highly sensitive to dispersion treatment, liquid partition and wall-interaction settings. These observations identify the mechanisms and modelling choices that most strongly influence the current separator predictions.

The findings must nevertheless be read skeptically. The open carrier-flow liquid balance, incomplete DPM trajectories and unmatched coupled snapshots prevent the current values from supporting quantitative separator-efficiency or causal wall-film claims. The report therefore establishes provisional model behaviour and a clear route for matched, mass-accounted follow-up studies rather than a validated performance result.

8. Future Work

Acknowledgements

Thank important people here. Be sure to thank your mum.

References

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Appendix A: The First Appendix

Appendix Text

Appendix B: Second Appendix

Some appendix text.